

Exploring sustainable energy transitions in sub-Saharan Africa residential sector: The case of Nigeria

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ABSTRACT

Energy system models can be used to explore future sustainable energy pathways for the household sector in order to inform policy decisions. However, few residential energy system models have been developed for the sub-Saharan Africa (SSA) region and they do not account for the key diversities of energy use within the sector. This paper addresses this gap by developing a new residential energy model for Nigeria—the most populous country in SSA. The model is then used to explore different energy transition scenarios for the sector. Results indicate that the final energy consumption of the rural households will reduce and that of the urban households will increase significantly by 2050 when compared to the base year levels. The results indicate that cooking remains the most energy-intensive end-use. Analysis of the scenarios reveals that realising Sustainable Development Goal number 7 in Nigeria will drastically reduce the final energy demand of the sector. The transition from fuelwood to liquefied petroleum gas (LPG) for cooking will reduce indoor air pollutants considerably but will increase CO₂ emissions by 2050. The results also indicate that including the cost of externalities in energy cost makes electricity and biogas more economically viable options for cooking. While the model results are insightful, the paper argues that robust and dedicated policies are needed for a sustainable energy transition of the Nigerian residential sector.

1. Introduction

Sub-Saharan Africa (SSA) is the central point of global energy poverty, yet its portfolio of energy resources is more than enough to satisfy its overall energy requirements if they are well developed and evenly distributed [1]. Presently, around 588 and 783 million persons in SSA do not have access to electricity and clean cooking fuels respectively [2]. The situation even becomes overwhelming in the context of climate change given the fact that the current energy system of most SSA countries is dominated by fossil fuels and traditional biomass [3]. Advancing socio-economic development in SSA will require significant investments in the sub-region's energy sector. Many SSA countries have realised this and have put energy at the top of their political agenda. Also, growing concerns about this issue have made the United Nations sustainable development goals (SDGs) to capture energy access, renewable energy, and energy efficiency as its goal number 7 (SDG 7)¹

Energy use in the household/residential sector plays an important role in the global energy system and in achieving some of the SDGs such as poverty eradication and gender equality [2]. In SSA, the residential

sector accounts for over 70% of the total energy consumption of the sub-region [2]. This energy is mainly consumed in the form of traditional solid fuels (animal dung, charcoal, coal, and fuelwood) for cooking and heating and this contributes significantly to ambient air pollution [4]. Providing all households in SSA with modern energy to reduce environmental pollution and energy consumption will require effective policy reforms. Apart from the lack of adequate finance, a key obstacle to facilitating sustainable energy development in SSA is the lack of proper information by SSA policymakers about the possible impacts of different energy policy pathways.

Taking a closer look on one of the countries in SSA, Nigeria is the most populous and largest economy in Africa, endowed with a large menu of energy resources, but only about 61% and 6% of its entire population have access to electricity and clean cooking equipment respectively [2]. The country is currently struggling to provide modern energy for all its citizens. Realising this ambitious goal will require the triangulation of policies, coordinated support, and strong political will from the government. However, the design of such policies for a successful energy transition needs to be informed by quantitative assessments which take into account the role of technologies towards

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¹ SDG 7 is discussed in greater details in Section 3.5.

List of abbreviations and nomenclature

CAGR	Compound Annual Growth Rate
EAC	Energy access scenario
ECN	Energy Commission of Nigeria
E_{pc}	Per capita electricity consumption
ETSAP	Energy Technology Systems Analysis Program
FMLHUD	Federal Ministry of Land, Housing and Urban development
FMP	Federal Ministry of Power
GDP	Gross Domestic Product
GEF	Global Environment Facility
GIZ	German Corporation for International Cooperation
IAMs	Integrated Assessment Models
IEA	International Energy Agency's
IPCC	Intergovernmental Panel on Climate Change

LPG	Liquefied Petroleum Gas
MDGs	Millennium Development Goals
MtCO ₂	Million tonnes of CO ₂
NBS	Nigerian Bureau for Statistics
NESP	Nigerian Energy Support Program
NPC	Nigeria Population Commission
NREEEP	National Renewable Energy and Energy Efficiency Policy
PJ	Petajoule
REE	Renewable and energy efficiency scenario
REF	Reference scenario
SCO	Society Cost scenario
SDGs	Sustainable Development Goals
SSA	Sub-Saharan Africa
TIMES	The Integrated MARKAL-EFOM System
UN	United Nations
UNDP	United Nations Development Programme

decarbonising the household sector and ensuring energy security.

Energy system models are tools which can be applied to explore the future energy pathways of a sector or region. There is no specific method for grouping energy system models. From a broader view, they are generally classified into two: bottom-up and top-down models [5]. The bottom-up models represent technologies in detail but ignores macro-economic system feedbacks and microeconomic decision making. On the other hand, the top-down models describe macroeconomic feedbacks in detail but uses aggregate parameters for technology development. The detailed review of energy system models is presented in Ref. [6].

While scanning the relevant literature, it was found that there exists a plethora of researches on household energy system using some of these models. However, most of these studies have been for developed and few transitioning economies [7–18]. Residential energy system has been modelled for some developing countries and for the SSA sub-region [19]. However, individual countries in SSA have received less attention over the years. The energy consumption of some households in Ethiopia; an SSA country has been modelled using the Long-range Energy Alternatives Planning System (LEAP) model [20]. In Nigeria, the residential sector has been modelled only as part of integrated assessment models (IAMs) [21–23]. IAMs do not normally describe the diversity of the energy use of the household sector in details. Other authors have tried to specifically model the Nigerian household energy system but they have not incorporated the key spatial heterogeneity and environmental emissions from the sector [24,25]. There is an obvious shortage of quantitative modelling research on residential energy transitions in SSA and in particular, Nigeria. Currently, there is no energy policy for the household sector beyond 2030 in Nigeria. There is also limited literature on the co-benefits of energy transitions in the household sector based on quantitative analysis.

The overall aim of this paper is to investigate the energy system implications of different sustainable energy pathways for the Nigerian residential sector using a bottom-up optimisation model which accounts for the spatial heterogeneity of the households. Specifically, this paper seeks to:

- develop a bottom-up energy system model for the Nigerian residential sector
- examine the future trend of energy consumption for different policy pathways
- examine the implication of different policy pathways on environmental emissions.

This paper contributes to the state-of-the-art in this field by presenting a new model (TIMES-Nigeria-Residential or TINIRES) for the Nigerian residential sector. TINIRES is based on the TIMES (The

Integrated MARKAL-EFOM System) modelling framework. It consists of six residential energy services and their key drivers for rural and urban households. The model is one of the few dedicated residential energy system models for an SSA country. This paper also illustrates an approach that is replicable for developing a residential energy model for another SSA country where there is a shortage of information on the dynamics of household energy transitions. Through scenario analysis, this study provides significant insights into the challenges and opportunities facing Nigeria towards achieving SDG 7. Beyond the current study, the TINIRES model can also be used to explore other scenarios that have not been analysed here. While this research is considered as a preliminary attempt, the case study presented here will provide necessary and useful knowledge to inform policy decisions regarding sustainable energy transitions in Nigeria.

2. Overview of residential energy situation in Nigeria

The energy demand of the Nigerian household sector consists of demand from rural and urban areas. Over the years, an increase in per capita gross domestic product (GDP), improved lifestyle, and increase in population have led to an increase in the energy demand of the Nigerian residential sector. The energy mix of the sector comprises traditional solid biomass (fuelwood and charcoal), electricity, kerosene, and LPG [24]. The total commercial energy consumption of the sector has been varying over the years and has increased at a Compound Annual Growth Rate (CAGR) of 1.93%, from 101 Petajoules (PJ) in 1990 to around 148 PJ in 2010 [26] (Fig. 1).

Households use energy to satisfy different services such as cooking,

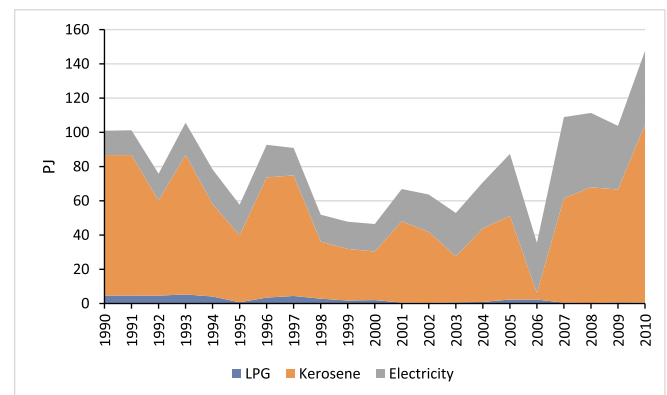


Fig. 1. Time trend of Nigeria residential sector commercial energy consumption.

lighting, heating and for operating appliances like refrigerators, fans, air conditioners, televisions (TVs) etc [25]. Electricity access in the rural area is still low with only about 41% of the households electrified [27]. Access to clean cooking equipment is even lower. Only about 3.5% of rural households have access to modern cooking fuels [27]. Few households depend on transition fuels like kerosene and the majority rely on traditional biomass for cooking. However, electricity access has improved substantially over the years in urban areas with 86% of urban households now electrified while only 8.5% have access to modern cooking fuels [27].

The energy requirements for cooking are diverse in the Nigerian residential sector. Fuelwood, charcoal, kerosene, LPG, and to some extent electricity are used for cooking [27]. Fuelwood is the predominant fuel for cooking especially in the rural areas of Nigeria [25]. It is usually collected from the forests around the villages in rural areas whereas, in the urban areas, they are purchased from the local vendors which sell them at a price that is very insignificant when compared to other cooking fuels like kerosene and LPG [25]. The high dependence on fuelwood for cooking has resulted in the depletion of many forests in Nigeria and has destroyed many natural ecosystems [28]. Moreover, the emission of air pollutants due to the combustion of fuelwood in inefficient stoves have severe health implications and contribute to around 79,000 deaths in Nigeria annually [29].

Fig. 2 shows the share of rural and urban households for different energy sources for cooking [27]. It may be observed that the use of modern fuels for cooking such as electricity and LPG are very marginal and can mainly be found in urban households. This can be attributed to the low-income levels of many Nigerian households which in turn limits their ability to climb the energy ladder [30]. However, other micro-factors such as availability and cultural norms influence the choice of cooking fuels [24]. Recently, there have been a lot of efforts and political discussions to provide all households in Nigerian with modern cooking equipment by 2030. Many actors (government and civil societies) have come into the 'game', but it is left to be seen how much of their good intentions will become a reality in the foreseeable future.

With respect to lighting, Nigerian households depend on electricity, kerosene, dry cell battery, candle, grass etc., but at the national level, electricity and kerosene constitute the main sources of lighting for the vast majority of households [27]. Electrical appliances such as refrigerators and air conditioners are mainly localised in urban households [24]. Most of the appliances found in Nigerian households are very old and inefficient. Some of them are imported from western countries after they have served their useful life² This indicates a large scope for energy saving within the sector. With the growing urbanisation and increased demand for a luxurious lifestyle, the use of refrigerators and air conditioners is expected to rise in the future.

3. Methodology

3.1. The energy system model

The overall methodological framework for this study is shown in Fig. 3. The TIMES model generator is employed in this study to build a residential energy system model for Nigeria. TIMES was developed in a collaborative program under the guidance of the Energy Technology System Analysis Program (ETSAP) of the International Energy Agency (IEA) [31]. TIMES is multi-period bottom-up partial equilibrium model generator for the sectoral, local, national, multi-regional, or global energy system. It uses an optimisation algorithm and assumes a perfect foresight to provide a least-cost energy system based on several user-defined constraints (e.g. resource availability, technology penetrations, emissions cap etc). It is also one of the best-known energy

system modelling platforms and it has been applied in over 70 countries. TIMES has been used in several studies³ some of which encompasses the energy system implication of coal ban in Kazakhstan's residential sector [32]; analysis of rural household energy development in an African village [33]; and decarbonisation of the Chinese building sector [34].

TIMES uses an engineering and economic approach—two distinct but interdependent approaches that distinguish it from other energy system modelling tools. Additionally, TIMES allows the modelling of technologies in details at a well-disaggregated level. The key advantage of TIMES model is that it allows the development and analysis of numerous scenarios; both technological and non-technological. TIMES uses a linear programming system to achieve its key objective of minimising the total discounted cost of an energy system. The objective function of the TIMES model is shown in Eq. (1) [31].

$$NPV = \sum_{r=1}^R \sum_{y \in YEARS} (1 + d_{r,y})^{REFYR-y} \times ANNCOST(r, y) \quad (1)$$

where *NPV* is the objective function of TIMES (i.e. the net present value of the total cost of all regions); *R* the number of regions in the study area; *YEARS* the set of years in the model for which there are costs; *d_{r,y}* the discount rate; *REFYR* the reference year for discounting; and *ANNCOST* (*r, y*) the total annual cost in region *r* and year *y*.

TIMES is a data-intensive model and the user is required to provide data on the energy supply resources and technologies as well as on the available and future stock of energy demand technologies. To successfully develop a model based on TIMES modelling framework, the user is required to provide the following category of data: global assumptions, energy service demands, energy carriers, energy demand and supply technologies, and environmental emission factors. A more elaborate description and documentation of the TIMES model is given in Ref. [31]. Among the categories of data listed above, the energy service demands, or useful energy demands are the most important as they are assumed to be the model driver and thus will be described in detail in the next section. The global assumptions consist of the number of regions, modelling timeframe, and discount rate. The developed model is a single region model with 40 years modelling horizon (2010–2050). A discount rate of 10% is been assumed which is the same rate that was used by the Energy Commission of Nigeria (ECN) for Nigeria energy system [35].

3.2. Energy service demands

3.2.1. Drivers of energy service demands

The main drivers of energy use in the model are population (or its structure: number of households and its size), urbanisation, income, and electrification rate. Population census is usually conducted by the National Population Commission (NPC). The last census conducted in Nigeria was in 2006 and then, the population of Nigeria was estimated at around 140 million persons. For the future projection of population, data was taken from the United Nations 2019 medium variant population prospects [36]. In 2010, the total population of Nigeria was around 160 million persons and it is expected to increase to around 401 million in 2050. In the same timeframe, the urbanisation rate is expected to increase from 44% to around 70%. Data on characteristics of Nigerian households are usually collected by the National Bureau for Statistics (NBS) through household living standard measurement surveys. Statistical analysis of the NBS data shows that there is no significant change in the size of rural and urban households between 1999 and 2010 [37,38]. In this study, average household sizes of 5.9 and 4.9 are taken for rural and urban households respectively. The total number of households has been projected to increase from 15.3 to 14.2 million in 2010 to around 20.5 and 57.3 million in 2050 for rural and urban households

² <https://www.premiumtimesng.com/news/headlines/218472-investigati-on-inside-nigerias-toxic-tokunbo-trade.html> (accessed 10/12/2018).

³ See link for several applications of TIMES model: <https://iea-etsap.org/index.php/applications> (accessed 21/12/2018).

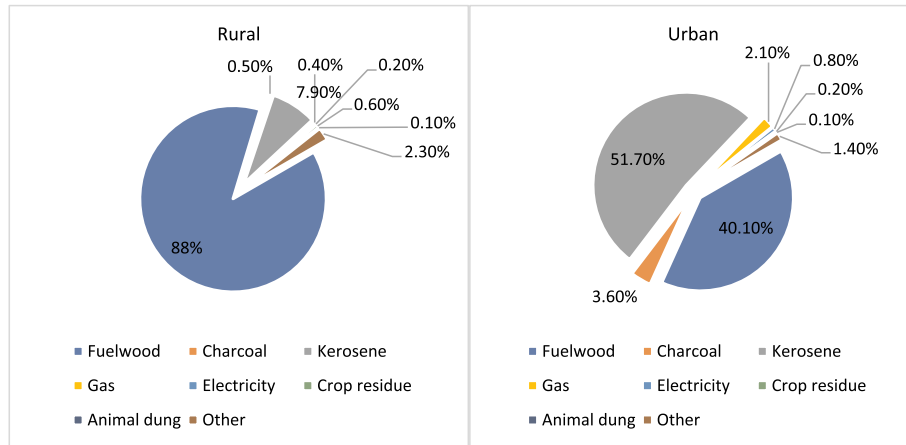


Fig. 2. Share of rural and urban households for different energy sources for cooking.

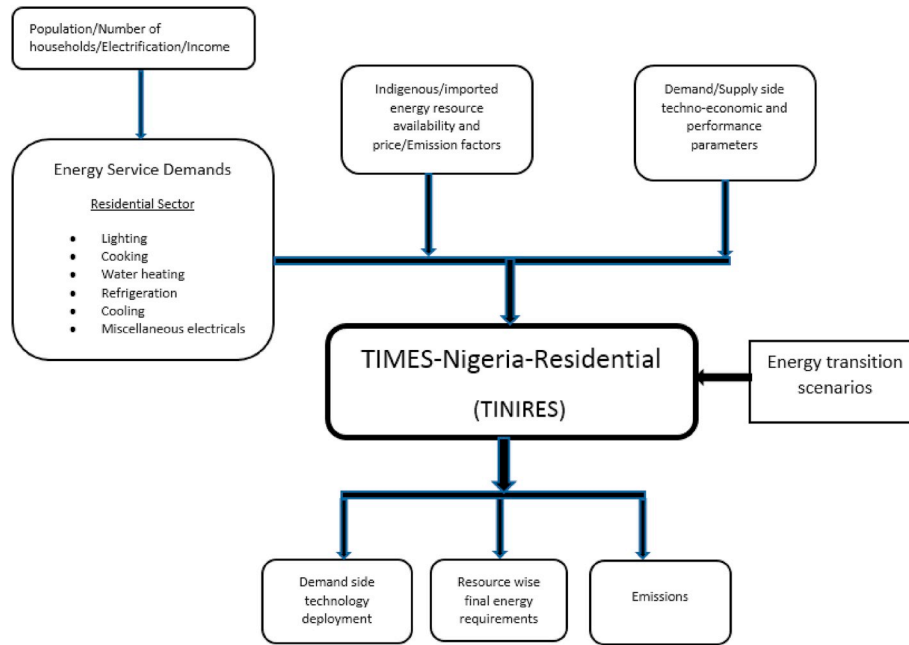


Fig. 3. Methodological framework.

respectively. Fig. 4(a) and (b) respectively, show the projections of Nigeria population and number of households.

Data on household income levels in Nigeria are scarce and this is the case in many SSA countries. Moreover, the household survey of the NBS rarely captures income levels of households. Therefore, per capita GDP has been used as a proxy for income—similar assumption has also been used in a global residential model where income levels were not available [39]. Data on Nigeria GDP (2010 constant price) was obtained from the reports of the NBS [40]. The monetary values were originally in Nigerian Naira (NGN), so it was converted to 2010 US\$ (US Dollars) using an exchange rate of US\$1 = NGN150, which was the average exchange rate in 2010. Nigeria GDP has been assumed to grow at an average annual rate of 7% from US\$ 364 billion in 2010 to about US\$ 4913 billion in 2050 (Fig. 5(a)). This is based on the 7.3% average per annum GDP growth observed between 2000 and 2015 [40,41]. Moreover, two earlier studies conducted by the ECN; Energy Implications of Vision 20:2020 [22] and the Nigeria Energy Calculator 2050 (NECAL2050) [35] both used 7% GDP growth per annum for the reference scenario. Consequently, the Nigerian per capita GDP has been projected to increase from US\$ 2284 in 2010 to around US\$ 12,315 in

2050 (Fig. 5(b)).

Electricity access is a key determinant of residential energy service demands especially in Nigeria where universal electrification is yet to be achieved. In Nigeria, electrification is mainly achieved by the extension of the national grid to households which do not have access to electricity. Consequently, electrification in Nigeria is in most cases driven by the policy of the government which is in turn determined by the economic growth of the nation. In this paper, the electrification rate is modelled using GDP as the determining factor for urban and rural households according to a logistic function [42] (Eq. (2)).

$$Elc_{r,u} = \frac{1}{1 + e^{-(k+GDPx_0)}} \quad (2)$$

where $Elc_{r,u}$ is the electrification rate for rural/urban households, while k and x_0 are the coefficients of the regression.

Gretl—an open-source econometric statistical software is used to

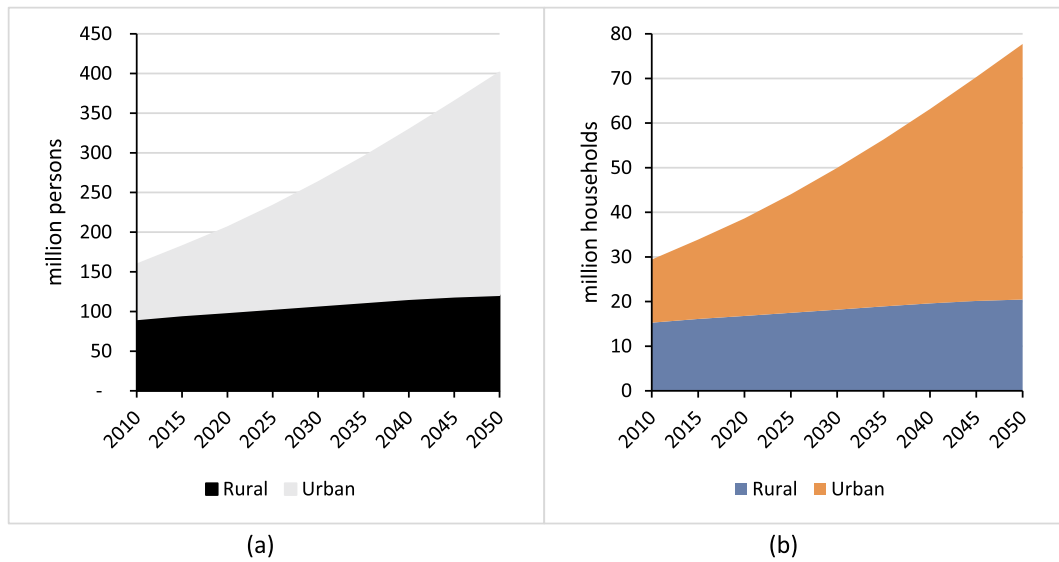


Fig. 4. Future projection of rural and urban population (a) and households (b) in Nigeria.

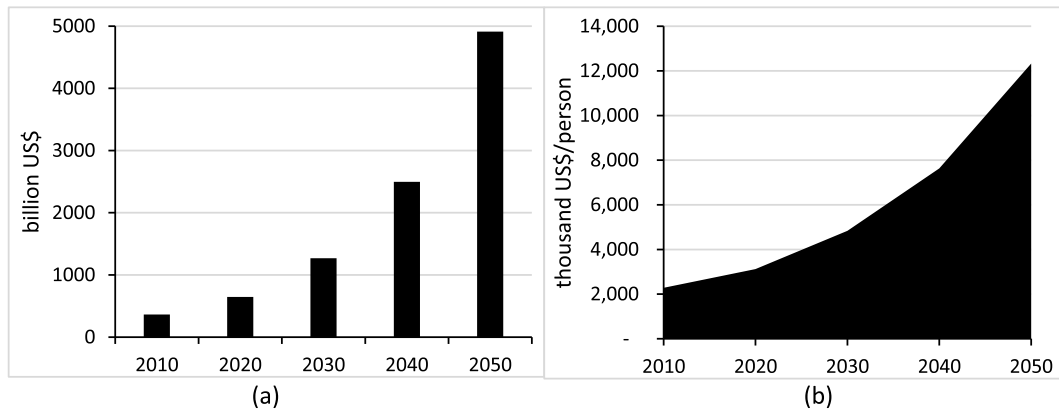


Fig. 5. Projection of Nigeria GDP (a) and per capita GDP (b).

determine the coefficients of the regression⁴ Historical data on rural and urban electrification rates were taken from the World Bank (WB) data [43]. The regression results show a good relationship between GDP and rural electrification with R^2 value of 0.69 ($k = -2.06$ and $x_0 = 2.32E-08$). The future projection shows that rural electrification rate will increase from 35% in 2010 to around 100% after 2040. However, no significant relationship was identified between GDP and urban electrification. To determine future urban electrification rate, the first author had a personal discussion with senior officials from the Nigeria Rural Electrification Agency (REA). In 2010, urban electrification rate was around 80% while 100% electrification rate was assumed by 2030 based on the discussions with officials of the REA. Fig. 6 shows future electrification rates for both rural and urban households of Nigeria.

With the above projections of residential energy service demands drivers, the useful energy demands of various end-use functions are computed. In the context of this study, six end-use functions have been considered for both rural and urban households of Nigeria and they are as follows:

- Lighting (LT)
- Cooking (CK)

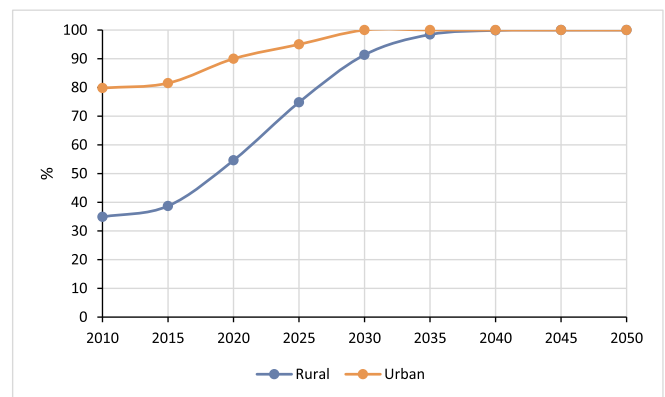


Fig. 6. Projection of Nigeria electrification rate.

- Water heating (WH)
- Refrigeration (RF)
- Cooling (CL)
- Miscellaneous electricals (MS)

⁴ See link for a detailed description of Gretl: <http://gretl.sourceforge.net/> (accessed 23/12/2018).

Energy service demand (ESD) of a particular end-use function can be determined by 3 variables viz, activity level (A), structure (S), and

useful energy intensity (I) as per Eq. (3) [44]. Here, the activity level refers to the main driver of residential energy demand. The structure represents other factors that determine energy demand, and useful energy intensity or specific energy consumption is the actual amount of energy consumed per unit of the activity while accounting for losses. This approach has been employed in this paper to determine the energy service demands or 'useful energy demands' of the key end-use functions.

$$ESD = A \times S \times I \quad (3)$$

3.2.2. Lighting

In this study, only electricity and kerosene are considered as the main sources of lighting in the country. It was assumed that all electrified households in Nigeria (rural and urban) depend on electricity for lighting, while the unelectrified households depend on kerosene. The useful energy demand for lighting (LT_{ESD}) for rural and urban households of Nigeria is estimated based on Eq. (4). The useful energy intensities for electric and kerosene lighting are taken as 454 kWh/household/year [45] and 1.22 GJ/household/year [25] respectively.

$$LT_{ESD} = HT_{r,u(y)} \times IL_{e,k} \times f_{el,nel} \quad (4)$$

where $HT_{r,u(y)}$ is the total number of households in year y (either rural or urban); $IL_{e,k}$ the specific energy consumption of electric or kerosene lighting; and $f_{el,nel}$ the fraction of the electrified or non-electrified households.

3.2.3. Cooking

Data on useful energy demand for cooking in Nigeria is sparse. However, the household surveys of Silviconsult [46] and Triple "E" [47] puts the useful energy demand for cooking between 700 and 900 MJ/capita/year. In this study, 800 MJ/capita/year has been assumed just as in Ref. [25]. Some studies have hypothesised that the energy requirement for cooking decreases as income increases due to the consumption of food outside the household [48]. However, another study has shown statistically that there is no significant relationship between income and cooking energy requirement [49]. In this study, the per capita cooking energy demand is considered to remain constant and the total useful cooking energy demand (CK_{ESD}) for rural and urban households is estimated through Eq. (5).

$$CK_{ESD} = Pop_{r,u(y)} \times IC \quad (5)$$

where $Pop_{r,u(y)}$ is the total population in year y (either rural or urban) and IC the useful energy intensity of cooking.

3.2.4. Water heating

Hot water is used for the bathing purpose in Nigerian households. Water heating energy service demand in the residential sector refers to the useful energy required to raise the temperature of water from one point to another desirable point and it primarily depends on the number of households and the climate of the region. Nigeria is in the tropics and the requirement of hot water for bathing is quite marginal. Thus, it is assumed that only 20% of the total households in Nigeria require hot water for bathing. Moreover, a similar assumption has been made for India—a tropical country in an earlier study on Indian national energy map [50]. Consequently, useful energy demand for water heating (WH_{ESD}) is estimated using Eq. (6) [51].

$$WH_{ESD} = HT_{r,u(y)} \times CW \times \Delta\theta \times m \times 365 \times f_{HWT} \quad (6)$$

where CW is the average daily water demand for bathing per household; $\Delta\theta$ the temperature change required (35 °C); m the specific heat capacity of water (4187 J/Kg°C); and f_{HWT} the fraction of households that heat water for bathing.

The initial and final temperatures of water are taken as 25 and 60 °C respectively [52], and the average household water demand for bathing

as 17.6 L/capita/day [53].

3.2.5. Refrigeration, cooling, miscellaneous electricals

A common characteristic of these end-use functions is that they are satisfied with electricity. The energy demand of these end-use functions is determined by the appliance penetration which is, in turn, a function of income, corrected by electrification rate as well as gradual efficiency improvements over time. There are different approaches to modelling appliance penetration [54]. However, the general trend of appliance penetration follows an S-shaped function such as the Gompertz and logistic functions [19]. In this paper, a simple logistic model which is appropriate for modelling penetration of appliances ($AP_{r,u}$) is used as per Eq. (7).

$$AP_{r,u} = \frac{\alpha}{1 + e^{-(k + Incx_1 + Elcx_2)}} \quad (7)$$

where α represents the saturation level of an appliance which we assume maximum as 100%; Inc and Elc represent income and electrification rates respectively; while k , x_1 and x_2 are the coefficients of regression.

An important aspect of modelling appliance uptake is to gather appliance penetration data as much as possible. Historic appliance penetration rates for rural and urban households are gathered from the household surveys of the NPC [37,55–57] and NBS [27,38,58–61]. Analyses show significant relationships between income and refrigerator as well as air conditioner penetration. However, for other miscellaneous electrical appliances, we were unable to gather historical data for all the appliances considered (TV, iron, computer, washing machine, fan, and DVD player). Thus, the penetration rate of TV which was available for both rural and urban households is used as a proxy for other electrical appliances. The regression outcomes are shown in Table 1.

Figs. 7–9 illustrate the penetration rates of refrigerators, air conditioners, and miscellaneous electrical appliances respectively.

To determine the useful energy demands of refrigeration, and air cooling, specific energy consumptions of 542 and 828 kWh/household/year respectively, were used [45]. For other miscellaneous appliances, their energy service demand will depend on the wattage of appliances and the hours of usage. Assumptions of average appliance wattage and usage norms in Nigeria have been adapted from Ref. [62] and they are shown in Table 2.

Through Eq. (8), useful energy demands for appliances have been obtained.

$$A_{ESD(RF, CL, MS)} = \frac{HT_{r,u(y)} \times IA_t \times AP_{r,u}}{ei} \quad (8)$$

where $A_{ESD(RF, CL, MS)}$ is the useful energy demand for either refrigeration, cooling, or miscellaneous electricals and IA_t the energy intensity of appliance t ; and ei the gradual efficiency improvement over time.

3.3. Technologies and energy sources

Appendix A illustrates a range of residential energy demand technologies that have been modelled to satisfy the energy service demands determined above. Single technologies that deliver multiple residential energy services are also factored in. For example, in rural Nigeria, traditional fuelwood stove is being used to satisfy cooking and water

Table 1
Regression results for electrical appliances.

		α	k	x_1	R^2
Rural	Refrigerator	0.70	−3.737	6.05E-06	0.647
	Air conditioner	0.40	−4.690	2.69E-06	0.970
	Miscellaneous	1.00	−3.593	7.50E-06	0.835
Urban	Refrigerator	1.00	−1.468	3.64E-06	0.707
	Air conditioner	0.75	−4.117	3.93E-06	0.971
	Miscellaneous	1.00	−0.993	5.71E-06	0.973

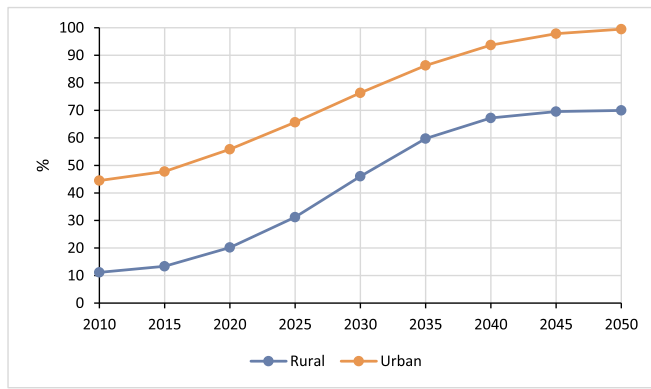


Fig. 7. Penetration levels of refrigerators in Nigerian households.

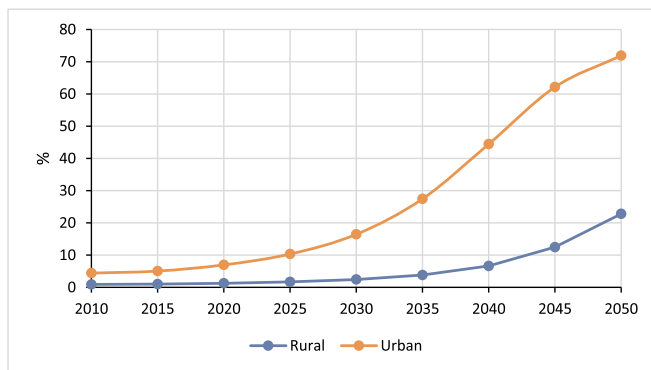


Fig. 8. Penetration levels of air conditioners in Nigerian households.

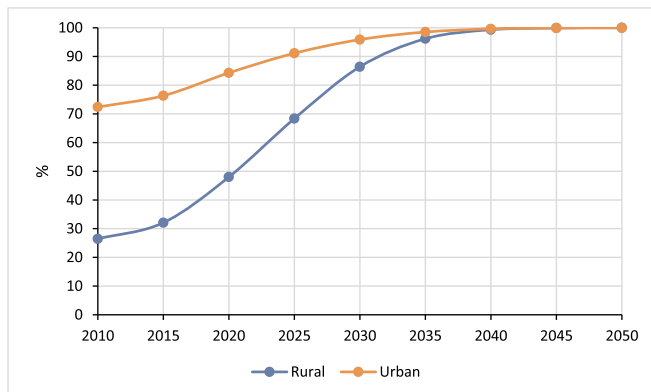


Fig. 9. Penetration levels of miscellaneous in Nigerian households.

Table 2
Appliance usage norms in Nigeria.

Electric appliance	Work hrs/day	Rating (Watt)	UEC (kWh/year)
TV set	3	120	131
Iron	1	1000	365
Laptop	8	100	292
Washing machine	1	1000	183
DVD player	3	20	22
Fan	7	75	192
Total			1184

heating demands. After specifying these demand technologies, the model estimates the final energy demand as well as fuels consumption using the least-cost fuel and technology (ies) combination. The supply

Table 3
Air pollutants emission factors.

Air pollutant	Fuel	Emission factor (kt/PJ)	Source
CO	Fuelwood	5.461	[66]
	Kerosene (cooking)	1.299	[66–68]
	Kerosene (lighting)	0.255	[69]
	LPG	0.442	[66]
NMVOC	Fuelwood	1.085	[70]
	Kerosene (cooking)	0.002	[70]
	Kerosene (lighting)	0.002	[70]
	LPG	0.003	[70]
NO _x	Fuelwood	0.300	[71]
	Kerosene (cooking)	0.030	[69]
	Kerosene (lighting)	0.030	[69]
	LPG	0.044	[72]
PM _{2.5}	Fuelwood	0.259	[73]
	Kerosene (cooking)	0.077	[74]
	Kerosene (lighting)	1.994	[75]
	LPG	–	
SO ₂	Fuelwood	0.034	[76]
	Kerosene (cooking)	0.009	[77]
	Kerosene (lighting)	0.151	[78]
	LPG	–	

side of the model consists of both renewable and non-renewable sources of energy. The urban households' electricity is assumed to be supplied by the national grid, rooftop solar PV, and gasoline/diesel generator sets. However, for rural households, technologies such as small hydro, gas turbines, and hybrid diesel-PV-wind system are included. Other fuels included in the supply side of the model encompasses fuelwood, kerosene, and LPG which mainly satisfy cooking energy service demand. Additionally, to develop an optimisation model, techno-economic parameters of the technologies need to be specified. Cost data on household appliances have been taken from Jumia⁵ and Konga⁶; two online shopping sites in Nigeria. Wherever local data is not available, data from the IEA ETSAP technology database⁷ South African TIMES (SATIM) database⁸ as well as experts' opinions were taken. Cost data on electricity supply technologies and grid electricity price in Nigeria were taken from Ref. [63], while the future prices of fossil fuels were gotten from the U.S. Energy Information Administration (EIA) annual energy outlook 2018.⁹

3.4. Environmental emission factors

To understand the implication of residential energy transitions on climate mitigation and air pollutants emissions, CO₂ and local air pollutants emissions are modelled: carbon monoxide (CO), non-methane volatile organic compounds (NMVOC), nitrogen oxides (NO_x), particulate matter (PM_{2.5}), and sulphur dioxide (SO₂). The standard IPCC emission factors for CO₂ emissions are considered. The study used emission factors of 68.61, 70.79, 72.60, 63.10, and 55.82 kilotonnes/petajoule (ktCO₂/PJ) for gasoline, kerosene, diesel, LPG, and natural gas respectively [64]. Nigeria grid emission factor is taken as 163 ktCO₂/PJ [65]. However, emission factors of different air pollutants will depend on the type of fuel and technologies used for cooking and lighting in Nigeria. While scanning the available literature, we realise the paucity of data on air pollutants emission factors in Nigeria. Thus, emission factors were taken from other countries that share similar cooking and lighting cultures/technologies with Nigeria. The calorific values of the

⁵ <https://www.jumia.com.ng/>.

⁶ <https://www.konga.com/>.

⁷ <https://iea-etsap.org/index.php/energy-technology-data>.

⁸ <http://www.erc.uct.ac.za/groups/esap/satim>.

⁹ <https://www.eia.gov/outlooks/aeo/>.

fuels were used to convert the emission factors to standard TIMES units (kt/PJ). The emission factors and their various sources are described in Table 3.

With the above information, the TIMES-Nigeria-Residential model (TINIRES) is developed. The simplified version of the reference energy system is shown in Fig. 10.

3.5. Scenarios description

The United Nations SDGs represent a significant milestone in the global policy framework as it encompasses all aspects of development, from poverty and hunger; to energy and climate change; to health and decent jobs etc. [79]. The SDGs are intertwined¹⁰ and thus, realising one of the goals might either have a positive or negative impact on another one. There have been international and local discussions around the SDGs and countries across the globe are called upon to align their socio-economic development goals with the 17 SDGs. While developing the scenarios; emphasis was laid on SDG 7 (energy)—an important element of socio-economic development that was missing in the millennium development goals (MDGs). The three targets of SDG 7 viz.

- 7.1 ensure universal access to affordable, reliable and modern energy services by 2030
- 7.2 increase substantially the share of renewable energy in the global energy mix by 2030
- 7.3 double the global rate of improvement in energy efficiency by 2030

have been captured while developing the scenarios. The federal government of Nigeria (FGN) has also shown its commitment to the realisation of SDG 7 in the country. In 2016, the FGN developed the Sustainable Energy for All Action Agenda. This document outlined the roadmap for achieving universal modern energy in the country and it was approved by the inter-ministerial committee on renewable energy and energy efficiency (ICREEE). Details of the action plan are given in Ref. [80]. It is believed that realising SDG 7 will go a long way towards contributing to the realisation of other SDGs. Moreover, a recent study has shown that the benefits of achieving SDG 7 outweigh the negative impacts on other SDGs [79]. Accordingly, the developed model is used to explore a reference scenario and other future energy transition scenarios for the Nigerian residential sector based on the targets of SDG 7. While the targets are for 2030, the analysis extended to 2050 in order to understand the ripple effects of the policies if they are maintained till then.

Reference (REF): This scenario is the most basic task in energy system analysis as it provides the benchmark for comparison with alternative policy scenarios. TIMES—an optimisation model provides the least-cost energy system, and this may not portray the reality in the Nigerian residential sector as some households do not often make energy choices based on the cost [81]. Therefore, bounds/constraints are also included on technologies uptake in the rural and urban households based on the current expectations and consistent with past trends as reported in the literature as well as stakeholders' consultation.

Energy Access (EAC): The target of SDG 7.1 is considered in this scenario. Access to modern forms of energy is central to human development as it enhances the integration of the poor households into the mainstream of a society which in turn, accelerates the realisation of many other SDGs. There is a clear nexus between energy access and poverty eradication (SDG 1). While access to modern forms of energy is not just enough to ensure poverty eradication, however, productive uses of energy for activities that generate income in sectors like agriculture and industry can go a long way to break the vicious circle of poverty in rural households. It may also be noted that in SSA countries like Nigeria, the burden of gathering fuelwood for cooking and the cooking itself is

usually borne by the women and children [82]. They spend an average of 1.5 h per day, gathering fuel for cooking. Additionally, the loads that they carry (around 25–50 Kg [83]) impacts their overall wellbeing negatively. Hence, providing modern energy access will go a long way to open new economic opportunities for women as they no longer spend time gathering fuelwood from the forest (SDG 8). The time saved can also be redirected towards education which in turn drives poverty eradication and improve gender equality (SDG 5) [84]. High reliance on traditional solid fuels for cooking also means that women are the ones most affected by the diseases attributed to indoor air pollutants. Access to modern energy forms will also improve health (SDG 3) as the number of deaths attributable to indoor air pollutants will be reduced. For this scenario, it is assumed that all households (rural and urban) will achieve 100% electrification by 2030 and kerosene lighting stops. Much of rural Nigeria is expected to be connected via off-grid systems by 2030, so the model is allowed to access other available technologies beyond the central grid. All households are assumed to shift towards LPG for cooking by 2030 which is in line with the National Renewable Energy & Energy Efficiency Policy (NREEEP) [85]. All other assumptions are kept the same as in the REF scenario.

Renewable Energy & Energy Efficiency (REE): This scenario investigates the targets of SDG 7.2 and 7.3. There is now scientific evidence on the impacts of energy generation and utilisation on the climate [86]. Application of renewable energy technologies and energy efficiency have been identified as key options to mitigate climate change (SDG 13) [87]. Another concept in energy development is energy security. Energy security refers to the continuous supply of energy resources at a price which is affordable¹¹ Renewable energy application will ensure that GHG emissions from the Nigerian fossil-fuel-based economy is reduced while energy efficiency will go a long way to reduce primary energy supply. Thus, to decarbonise the Nigerian energy system as well as ensure energy security, renewable energy and energy efficiency will play significant roles [88]. For this scenario, renewable energy and energy-efficient technologies, as well as advanced technologies, are introduced in the Nigerian residential sector. The NREEEP was studied to ensure that the assumptions and storylines used are in line with the Nigerian government policies for the residential sector. We were also conservative in the analysis in order to avoid unrealistic scenarios. Induction, improved wood, and biogas stoves are moderately introduced in addition to LPG stove in rural and urban households. Solar water heater (SWH) and advanced technologies for cooling, refrigeration among other energy services are also introduced. Other assumptions remain the same as in the EAC scenario.

Society Cost (SCO): To determine the technologies that would most benefit the Nigerian households, it is important to consider the true cost of the technologies. Society cost is based on the cost of externalities (environmental or health) borne by the society which can be attributed to emissions due to an energy technology. These kinds of costs are not reflected in the capital costs of the technologies as they have no direct financial implication on the consumer. Based on life cycle assessment¹² studies have shown that the most important externalities cost can be derived from the impacts of climate change due to GHG emissions coming from the combustion of fossil fuels as well as the health impacts of indoor air pollutants as in the case of inefficient biomass cookstoves. It is worthwhile to note that there are other externalities costs due to energy production such as land degradation, water pollution, system integration costs, etc. However, the focus of this study is on the costs of CO₂ emissions and air pollutants. Over the last two decades, there have been several efforts to express externalities in monetary value but none has been done so far for Nigeria [63]. While assessing the externalities costs of energy production using a technology, it is important to note that costs will vary for different countries and local circumstances. For

¹⁰ <https://sustainabledevelopment.un.org/sdinaction/2016report>.

¹¹ <https://www.iea.org/topics/energysecurity/whatisenergysecurity/>.

¹² http://www.externe.info/externe_d7/.

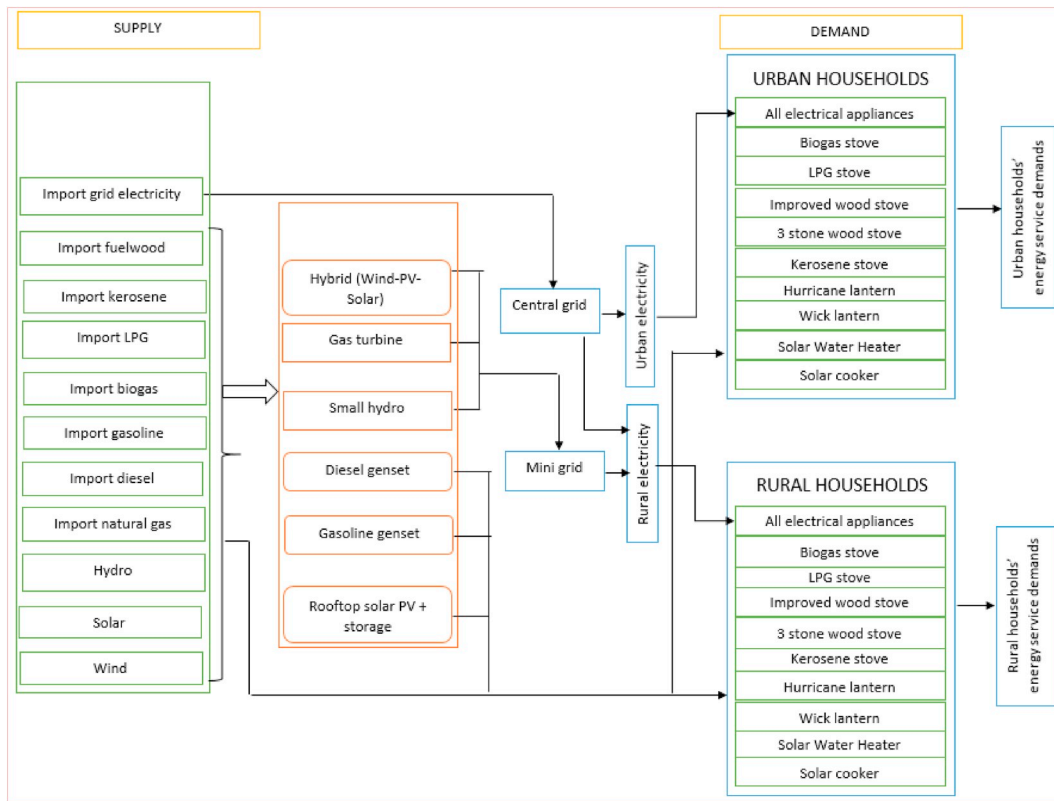


Fig. 10. Simplified structure of TINIRES model.

instance, the externalities cost will depend on the emission factors of the technology which is in turn, a function of the fuel types and technologies configurations [89]. For the case of air pollutants, it will depend on household exposure to indoor air pollutants which also depend on the type of ventilation system in the house and how the health implication is valued in the country. Owing to the non-availability of Nigerian-specific data, this study mainly relies on foreign data. The IPCC has mentioned US\$/tCO₂ 17 (lower value), 90 (best guess), and 350 (high value) for externalities costs of CO₂ emissions [90]. In this paper, the best guess (90 US\$/tCO₂) is used for the analysis. Data on externalities costs of air pollutants considered in this study were taken from an earlier study on rural residential energy use in a South African village and they are given as follows [33]: CO (US\$/t 946), NMVOC (US\$/t 339), NO_x (US\$/t 88), PM_{2.5} (US\$/t 339), and SO₂ (US\$/t 713). It is worthwhile to note that these costs are only indicative and as such, should be considered with a marginal view. In this scenario, the impact of the externalities costs on residential sector energy demand and emissions is experimented. While doing the same, all constraints are removed from the model except for solar cookers (not an appropriate technology for Nigerian residential sector) and all households are allowed to move towards the least-cost technologies to satisfy their energy service demands.

The detailed assumptions and numerical values characterising the scenarios are provided as a supplementary file to this paper (Appendix B). It is important to note that all the scenarios are based on the simplified assumption that demographic projections and technologies cost do not change over the modelling timeframe.

4. Model results

4.1. Final energy demand

4.1.1. Rural and urban energy demand

In the REF scenario, only the current technologies are available for the model and the results show that energy demand of the Nigerian

residential sector (rural and urban) will increase significantly in the future owing to the projected increase in income and number of households. Fig. 11 shows the final energy demand for different energy

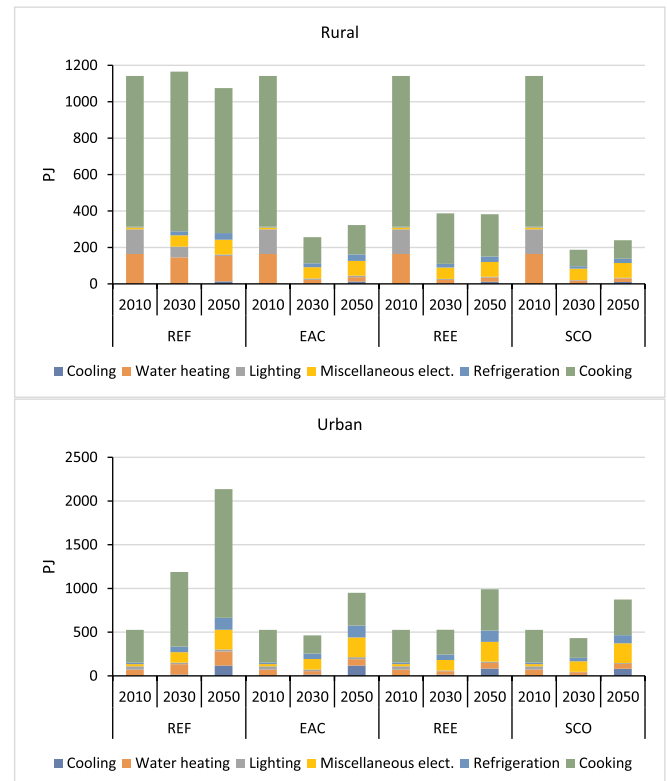


Fig. 11. Households final energy demand.

services in 2010, 2030, and 2050 for both rural and urban households. The total final energy demand of the entire household sector (rural and urban) grows from 1668 PJ in 2010–3211 PJ in 2050, increasing by approximately 100% in 40 years. The results show that the growth pattern of energy demand varies across rural and urban households. The final energy demand of the rural households is expected to decrease at a CAGR of 0.15%, from 1142 PJ in 2010 to 1075 PJ in 2050. It is worth noting that the energy consumption of the rural households account for about 68% of the total residential energy demand in 2010. However, in 2050, the share of the rural households in the total residential energy demand decreased to 33%. This is because of the projected urbanisation. Additionally, as the rural households gradually get richer, they tend to switch to more energy-efficient appliances and this, in turn, reduces the overall energy demand of the rural dwellers. However, in urban households, an opposite trend of energy consumption is seen when compared to rural households. As expected, with the influx of migrants from the rural areas in the search of economic opportunities, the energy consumption of the urban households increases significantly. The results indicate that energy consumption of the urban households will grow at a CAGR of 3.57%, from 526 PJ in 2010 to about 2136 PJ in 2050.

In the EAC scenario, all households switched to LPG for cooking in 2030, universal electrification is achieved in 2030 while the standard electrical appliances still remain in use in all households. Results show the total final energy demand in 2050 to be around 1273 PJ, a saving of about 60% in comparison to the REF case. It may also be recalled that the only changes done in this scenario are on the cooking and lighting technologies. This result helps us understand the huge potential of energy savings from the cooking sub-sector. The analysis for rural and urban energy usage in this scenario shows a similar trend with the REF scenario. Energy consumption of the rural households falls by a CAGR of 3.11% down to 323 PJ in 2050, which is about 70% reduction compared with the REF case. However, in urban households, energy demand is seen to increase at a CAGR of 1.49%–950 PJ in 2050. Despite the projected growth, analysis shows that urban households energy demand will decrease by around 56% in comparison to the REF scenario.

For the REE scenario, renewable energy technologies such as SWH and improved wood stoves as well as advanced technologies (from 2030) were introduced in the Nigerian residential sector. Results (Fig. 11) show that the total residential sector energy demand will rise to 1373 PJ in 2050 which is about 57% reduction when compared to the base or REF scenario. It may be noted that energy demand savings in this scenario are lower than the one achieved in the EAC scenario. This is because some households in the rural and urban areas depend on improved wood stoves for their cooking demands in this scenario which are not as efficient as the LPG stoves. Analysis on the categories of households shows that energy consumption of the rural households will decrease at a CAGR of 2.70% down to 382 PJ by 2050, which is around 64% reduction in comparison to REF scenario. In the same year, urban households' energy consumption increases at a CAGR of 1.60%–991 PJ; about 54% saving relative to the REF scenario.

In the SCO scenario, all constraints are removed from the model and an experiment was conducted to see how the model will behave when externalities cost are added to the cost of energy. The results of the SCO scenario showed some drastic changes in energy consumption. Firstly, the total energy consumption of the residential sector is seen to drop to 1112 PJ by 2050 which is also about a 65% reduction in comparison to the REF case. Across all scenarios analysed, the SCO scenario gives the maximum energy demand reduction. In the rural and urban households, energy demand decreases and increases at a CAGR of 3.83% and 1.27% to 239 and 873 PJ in 2050 respectively. These results correspond to a reduction in energy consumption by around 78% and 59% in rural and urban households respectively, when compared to the REF case.

From the three scenarios analysed now, it may be observed (Fig. 11) that the energy demand of urban households gradually reduces up to 2030 when compared to the base year (2010) levels but continues to increase from then up to 2050. This is because, despite the energy

access, renewables, and energy efficiency policies by 2030, other energy demand factors such as appliance penetration and the number of households continue growing even after 2030, and thus, overshadows the impacts of these policies on energy demand reduction. These results also suggest that there is a need for Nigeria to start focusing on household energy development policies beyond 2030.

4.1.2. End-use functions

Fig. 11 shows the projected energy consumption of the various end-uses considered in this study. As expected, cooking consumes the highest amount of energy across all categories of households in all the scenarios analysed. In the base year of the REF scenario, cooking accounts for 72.5% of the rural household energy consumption. This is followed by water heating (14.4%), lighting (11.9%), miscellaneous (0.8%), refrigeration (0.4%), and cooling (0.04%). By 2050, the share of cooking in the final energy demand in rural households slightly drop to 71%. In urban households, a similar pattern of energy consumption is observed as in rural households where cooking consumes the largest portion of energy. This further confirms the fact that cooking is the most energy-intensive activity in the Nigerian residential sector. However, due to several demand management measures and fuel efficiency enhancements in the demand technologies in the alternate scenarios, the share of cooking is seen to drop significantly by 2050. Model results show that the share of cooking in 2050 will be around 46%, 57%, and 38% in EAC, REE, and SCO scenarios respectively, in both rural and urban households. The analysis further shows that demand for cooling and miscellaneous electrical appliances will increase significantly in the urban areas and will consume more energy than water heating by 2050 across the alternative scenarios.

4.2. Fuel consumption and technology selection

4.2.1. Fuel consumption

Fig. 12 shows the total amount of energy consumed by different

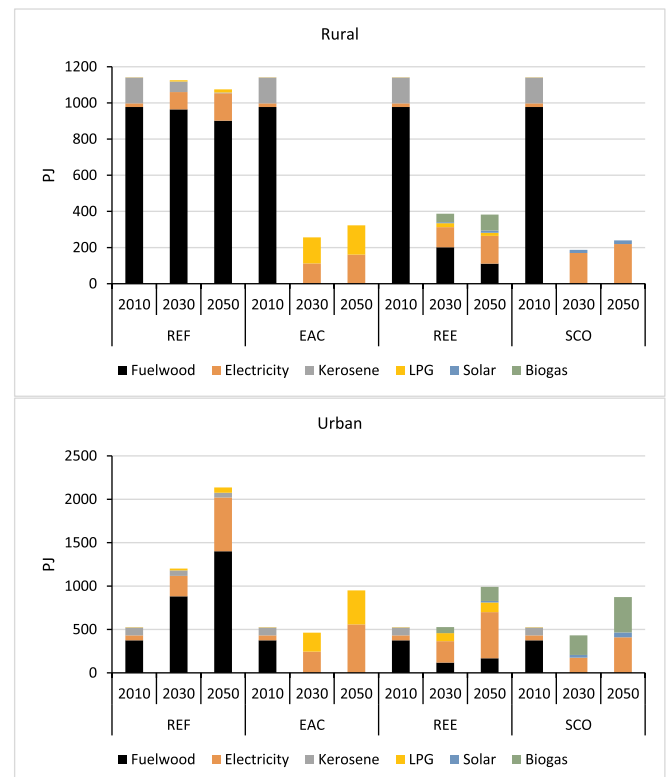


Fig. 12. Households consumption of energy carriers.

energy carriers in both rural and urban households. Predictably, results for the REF scenario show that fuelwood which is used for cooking and water heating continues to be the dominant source of energy especially in the residential sector where it accounts for about 86% and 82% of the total rural household fuels consumption in 2010 and 2050 respectively. For urban households, the share of fuelwood is lower when compared to the rural area. Fuelwood accounts for 71.3% of total fuel consumption in the base year and this value reduced to 68% in 2050. It is also observed that kerosene is the second most-consumed fuel in the base year across the two categories of households. However, by 2050, the consumption of electricity overtook that of kerosene and thus, accounts for the second-largest share of fuel use. Results show that LPG is the least consumed fuel in both rural and urban households.

In the EAC scenario, fuelwood remains the dominant fuel in all households until 2030 when it was substituted with LPG. Consequently, LPG and electricity remain the only fuels consumed by the residential sector after 2030. LPG is used for cooking and heating, while electricity is mainly used to satisfy other energy services. LPG consumption in rural households increases from 1 PJ in 2010 to 162 PJ by 2050, while electricity consumption increases from 20 PJ to 161 PJ in the same years. In the urban areas, LPG and electricity demand increase from 4 PJ and 57 PJ to 393 PJ and 558 PJ in 2010 and 2050 respectively. It is useful to note that electricity is the most consumed fuel in this scenario in 2050. Analysis of the REE scenario shows that beyond 2030, fuelwood, electricity, LPG, biogas, and solar will all become part of the household fuels. Electricity is the most consumed fuel in this scenario and accounts for around 43% and 55% of the total fuel consumptions of rural and urban households respectively by 2050. In the SCO scenario, cost was applied to CO₂ and air pollutants and the results show that electricity, biogas, and solar are the only consumed fuels in rural and urban households. Projected energy consumption of different energy carriers is detailed further in Fig. 12.

4.2.2. Technology selection

With respect to demand technology choices in the SCO scenario using the true cost of energy, the model selected induction and biogas stoves as the cost-optimal cooking technologies in the rural households while it picked only biogas for the urban households. For water heating, SWH and super-efficient geysers are selected for both the rural and urban households. Though the supply of electricity is not carbon-free, induction stove has been selected by the model because it does not release any local air pollutant during usage. Similar inference can be made for the biogas stove. For lighting services, LED is seen to dominate across the modelling horizon owing to its relatively higher efficiency over other lighting technologies. The analysis also shows that the super-efficient air conditioners are cost-optimal for the rural households not until 2030 when only the efficient air conditioners dominate. For the urban households, advanced air conditioners are seen to be the only selected technologies that are employed for cooling and similar technology is also selected for refrigeration for the two categories of households. These model choices can be attributed to the lower electricity consumption of super-efficient appliances. From the supply side, the model selected hybrid diesel-PV-wind, rooftop solar PV and small hydro as the best option for rural electrification. Hybrid diesel-PV-wind seems to be economically viable in the initial years, but small hydro appears to be the cheapest option in the long run where it supplies about 70% of the total rural electricity in 2050. The later choice of small hydro could be because of its construction time. However, for urban households, a combination of grid power supply and rooftop solar PV is seen to be the best option. In the initial years, the grid electricity supply is viable for about 98% of urban households, but this share reduced by 2050 when it is viable for about 52% of the households.

4.3. Per capita electricity consumption

Per capita electricity consumption (E_{pc}) has been identified as one of

the indicators for human development and general welfare¹³. Fig. 13 illustrates the results of residential per capita electricity consumption for each scenario. It may be observed that the E_{pc} of the Nigerian residential sector increased significantly in 2050 across each scenario in both rural and urban households when compared to the base year levels. Predictably, in the rural areas, E_{pc} in the REF scenario is expected to increase from 62 kW-hours (kWh) in 2010 to about 411 kWh in 2050, which is more than 5 times increase compared to the base year. In the EAC, and SCO scenarios, E_{pc} will rise by 6.4%, and 44.7% respectively, by 2050 in comparison to the REF case.

However, E_{pc} is seen to rise by only 1.4% in the REE scenario relative to the REF case. For urban households, E_{pc} in the REF scenario will rise from 227 kWh in the base year to around 585 kWh in 2050 or 157% increase when compared to the base year value. In the EAC scenario, E_{pc} in 2050 is expected to decrease by 9.9%. However, in the same year, the E_{pc} will fall by 14.4% and 40% in REE and SCO scenarios respectively, when compared to the REF scenario. From the numerical results, it may be observed that the per capita electricity consumption of the urban households is more than that of the rural households in all scenarios and in all years except for the SCO scenario. This is because of the induction stove that was selected in that scenario by the model as the main cooking technology for rural households. Thus, the current result also concurs to the assertion that there is a dichotomy of energy use between rural and urban households in SSA.

Another significant insight from the results which can be seen in the urban households is that the E_{pc} of the scenarios in which energy-efficient and renewable technologies are available for the households (REE and SCO) is lower compared to the REF case. However, it may be

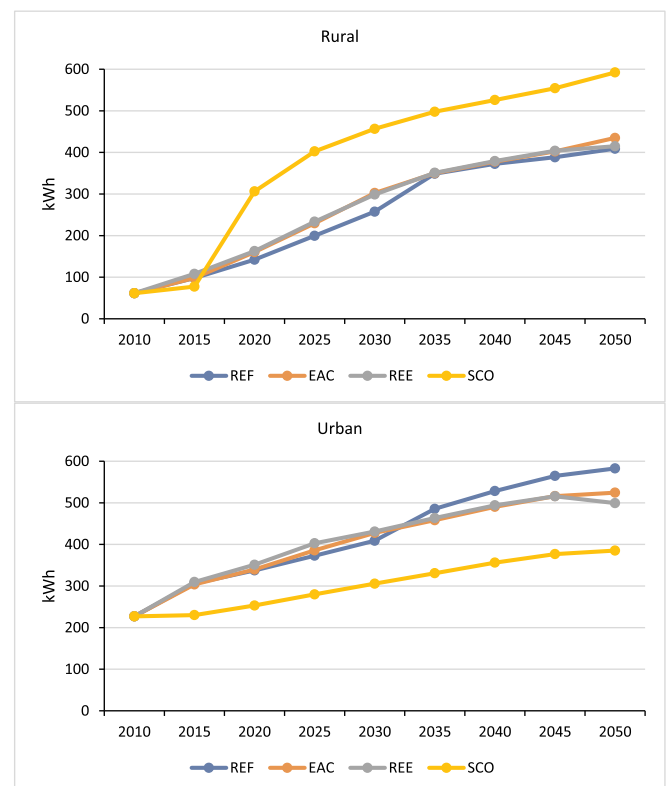


Fig. 13. Nigeria residential sector per capita electricity consumption.

¹³ <https://www.cgdev.org/media/electricity-consumption-and-development-indicators> (accessed 21/12/2018).

recalled from Section 4.2, that the percentage share of electricity consumption was maximum in the SCO scenario. With this in mind, it may be argued that E_{pc} is not a good indicator for human development in the context of low carbon development as it only focuses on the amount of electricity consumed by a country without considering energy efficiency and other sustainability issues. Thus, there is a need for better metrics in

explaining the role of electricity in human development.

4.4. Environmental emissions

4.4.1. Local air pollutants

In the REF scenario, for rural households, CO, NMVOC, NOx, PM2.5,

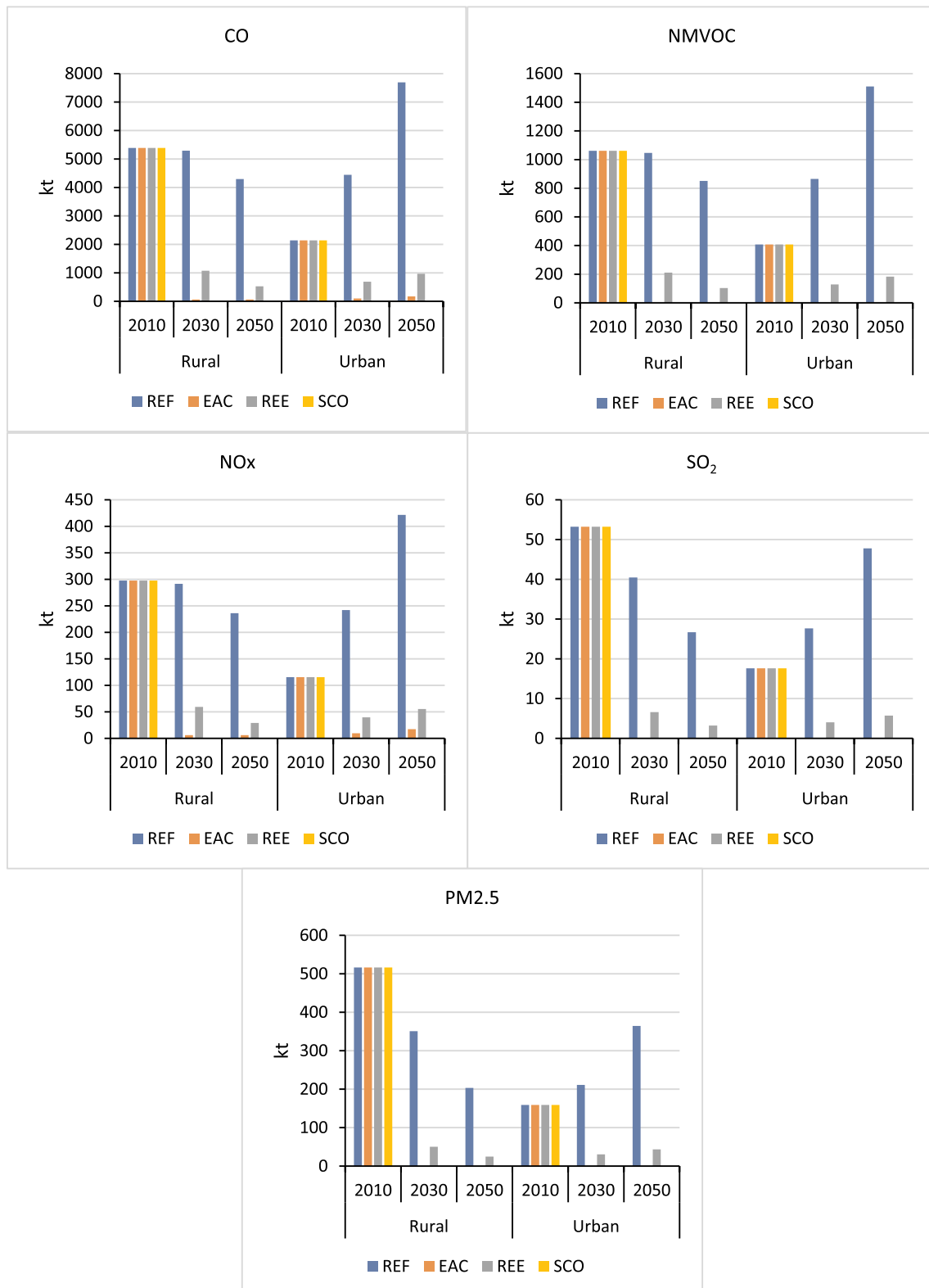


Fig. 14. Household air pollutants emissions.

and SO₂ emissions are projected to reduce respectively, by 21.4%, 20.8%, 21.6%, 62.3%, and 49.9% in 2050 in comparison with the base year levels (Fig. 14). This is due to the reduction of the number of households in the rural areas and the small penetration of efficient cooking and lighting devices in rural households. However, it is a reverse case for urban households. In 2050, CO, NMVOC, NO_x, PM_{2.5}, and SO₂ emissions are projected to increase by around 2.6, 2.7, 2.7, 1.3, and 1.7 times respectively, compared to the base year levels. This is due to the growing number of urban households. Despite the small penetration of efficient cooking devices in urban households, the growing number of households overshadows its impact on air pollutant emissions.

In the energy transition scenarios, the results show a drastic reduction of emissions compared to the REF case in both rural and urban households. In the EAC scenario, there is almost zero emission of NMVOC and PM_{2.5} after 2030, due to the switch to LPG and electricity for cooking and lighting services respectively. In the REE scenario, emissions also reduced but not as much as the EAC case when compared to the REF scenario. Analysis of the SCO scenario reveals there will be approximately zero emissions for all kinds of air pollutants as electricity and biogas are the only fuels employed in this scenario which is considered here to be relatively clean in terms of direct emissions. However, when air pollutants emissions from the power sector are accounted, it may become significant.

4.4.2. CO₂ emissions

Fig. 15(a) and (b) show the direct and indirect CO₂ emissions (including emissions from electricity generation) respectively, from the entire residential sector (rural and urban). The key sources of CO₂ emissions from the residential sector are kerosene and LPG which are mainly used for cooking, lighting for non-electrified households, and water heating. Household direct CO₂ emissions (Fig. 15(a)) in REF scenario decreases at a CAGR of 1.6%, from 16.7 million tonnes of carbon dioxide (MtCO₂) in 2010 to about 8.7 MtCO₂ in 2050 due to the use of electricity for lighting and water heating. In the EAC scenario, CO₂ emissions reach 35.1 MtCO₂ by 2050. This indicates a rise by about 3 times in comparison with the REF case. The increase is mainly coming from cooking where LPG serves as the only energy source in the residential sector after 2030. Though LPG is considered to be a clean fuel, this is only from the perspective of local air pollutant emissions and it is important to note that LPG is still a fossil fuel which releases GHGs during combustion such as the cooking process.

In the REE scenario, CO₂ emissions are seen to fall to 8.1 MtCO₂ in 2050 which is lower than the REF case and much lower than the EAC case. This is because of the penetration of renewable energies such as improved wood and biogas stoves in the cooking sub-sector. In this paper, biomass is considered to be a net-zero CO₂ emitter. However, in reality, biomass might not be a renewable source of energy especially when it is harvested unsustainably [19]. In the SCO scenario, results show that there will be no direct CO₂ emissions from the residential sector owing to the non-availability of fossil fuels in the sector. From Fig. 15(b), we may observe that including emissions from electricity generation in our results increases the total CO₂ emissions in all scenarios.

5. Discussion and options for the future

5.1. Discussion

Analysis of the model results shows that there are significant differences in energy use between rural and urban households. Thus, accounting for this heterogeneous feature of the Nigerian household sector helps us to understand the impact of different policies on energy use and environmental emissions across the two classes of households. It is not easy to make a comparison with previous studies as quantitative analysis on residential energy use is very limited for Nigeria. However, Ref. [24] have attempted to project Nigeria residential energy demands up to 2050 using the LEAP model. Comparing this research to his work shows around 49% and 8% variations of residential energy consumption in 2010 and 2050 respectively, as a result of the different methodologies used in the two studies. Projecting future energy demand using bottom-up methodologies such as TINIRES provides insights on the impacts of technologies on energy consumption [51]. This assertion is further confirmed in this study via the energy transition scenarios where efficient technologies were introduced into households. Beyond energy demand, bottom-up methodology also allows us to assess the impacts of certain technologies on GHG and air pollutant emissions.

In the REF, EAC, and REE scenarios, at least small amounts of electricity were consumed for services such as lighting and water heating. However, with the incorporation of externalities cost in the SCO scenario, there is a massive shift to electricity for most household services. The potential of electricity towards reducing household environmental emissions and improving the overall welfare of households explain why SDG 7 seeks to provide universal electricity for all by 2030. To a large

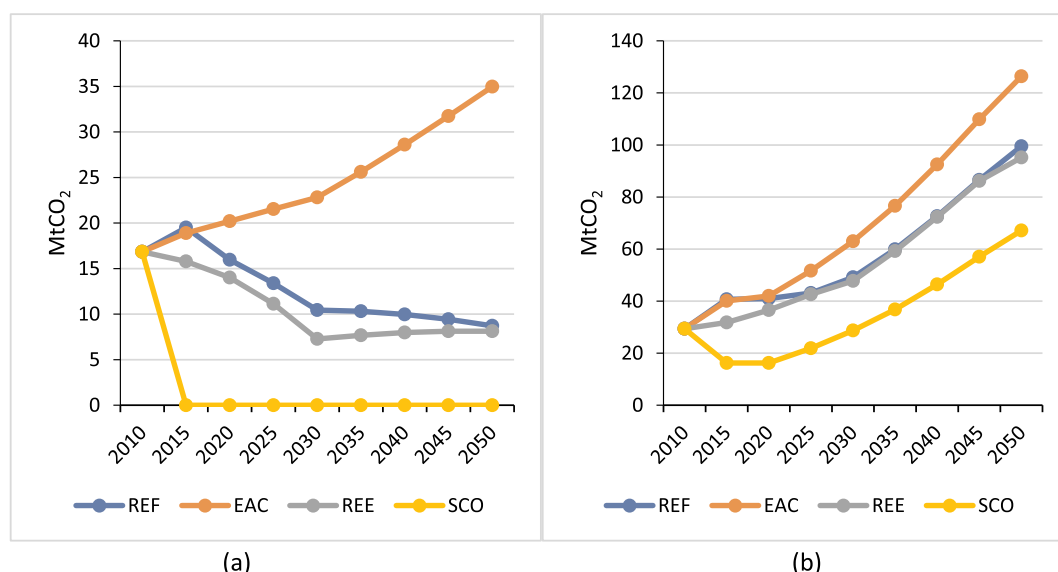


Fig. 15. Total household CO₂ emissions (a) direct (b) indirect.

extent, the result depends on how the cost of externalities will be valued in Nigeria. It may be recalled that the externalities costs applied in this study are only indicative values and further work is required in this area for the results of this model to be considered valid in the context of low carbon transition. This analysis is just to have a preliminary estimate and more accurate data as well as sensitivity analysis is required for long-term perspectives.

The technology-rich bottom-up model (TINIRES) developed in this study can further be improved by incorporating socio-economic heterogeneity among the rural and urban households whenever such kind of data is available. While estimating useful energy demands, the drivers of household energy consumption used in this paper are the ones that are fully understood. Hence, other factors beyond cost such as cultural preference which also affects household energy demand need to be studied more. This would allow us to test the hypothesis that other factors beyond income affect the choice of household energy use, using the newly developed TINIRES model [91]. However, the kind of data required for incorporating this type of hypothesis in TINIRES does not presently exist for Nigeria. We suggest going further with this research by investigating the economic implications of the energy transition scenarios. While achieving SDG 7 in Nigeria will significantly reduce household energy consumption and emissions, implications on employment/job creation or its rebound effects are unknown. More research is needed in this area using other kinds of models such as the computable general equilibrium and agent-based models.

5.2. Options for the future

For the future, the model results for the 3 alternative energy transition scenarios indicate that energy efficiency potential is very high in the Nigerian household sector, and can serve as the stepping stone towards low carbon energy development of the sector. The model shows that improvements in energy efficiency are the least-cost option for decarbonising the Nigerian household sector since it pays for itself from the reduced costs of energy. Nigerian planners need to promote LED as well as other energy-efficient bulbs while ensuring that incandescent bulbs are phased out of the country as soon as possible. It may be noted that most air conditioners and refrigerators in use in Nigerian households are usually imported from the western countries where they have already served their useful lives. This, therefore, calls for the promotion of energy efficiency standards for key appliances in the sector such as the 'top runner' program in Japan; where the most efficient appliance model in the market is set as a standard for future appliances [92]. Then the big question: why is the penetration of energy-efficient appliances low in the Nigerian household sector despite the relatively lower life-cycle cost? This is mainly because of the high initial capital cost of energy-efficient appliances. Hence, Nigerian policymakers need to provide targeted subsidies for these appliances for poor households and other incentives such as access to finance and soft loans as key strategies to encourage the deployment of energy-efficient appliances.

In the REE and SCO scenarios where SWHs were introduced for water heating, the model shows that it is the cheapest option for water heating services and the best alternative for reducing CO₂ and air pollutant emissions. The most important challenge for introducing SWHs in Nigerian households is the relatively perceived high upfront cost of the technology and lack of awareness. Efforts are underway to deploy SWHs in Nigerian households such as the pilot SWH project of the Nigeria Energy Support Program (NESP), planned by the German Corporation for International Cooperation (GIZ) in collaboration with the Federal Ministry of Power (FMP) and the Federal Ministry of Land, Housing and Urban Development (FMLHUD) [93]. While the proposed programs are essential, the future prospects of SWHs in Nigerian households strongly depends on the political will of the government to implement robust financing schemes for renewable energy technologies.

The model results suggest that moving all households to LPG for cooking by 2030 will significantly drop energy consumption and air

pollutant emissions, however CO₂ emission increases. Thus, policy-makers need to understand that there is a need for a trade-off between modern energy access and climate mitigation. The current National Renewable Energy and Energy Efficiency Policy [85], proposes moving all households to LPG by 2030 which can be partially attributed to the abundance of oil in the country. However, there is a need to reconsider this policy. The results suggest that biogas and electricity are the best options when the true cost of using LPG is included in the context of climate change. While the model results are significant, it is also important to consider the institutional dimensions of the results. Electricity access for basic household services like lighting is a major problem now and the Nigerian government might not consider the use of electricity for cooking as a priority issue. Thus, there is a need to tap into the full menu of biogas resources in the country for cooking purpose. Nigeria has a biogas potential of around 25.53 billion m³ of biogas per annum which is more than enough to satisfy the cooking demands of Nigerian households [94]. This will also go a long way to decarbonise the residential sector. Biogas technology development in Nigeria has remained at the scale of laboratory experiment and few pilot schemes such as the 425 L biogas plant built at the Usman Danfodiyo University Sokoto [94]. This is because of the widespread ignorance of the benefits of this technology and lack of an adequate policy framework for harnessing it in the country [94].

The model result also indicates that when the true cost of energy production is considered as per the SCO scenario, small hydro will become a major player in rural electrification. Currently, out of the 3.5 GW of small hydro potential in the country, only about 0.064 GW is being exploited [35]. The REA has laid emphasis on solar PV for decentralised rural electrification. While this option, as well as the hybrid diesel-PV-wind system, are already cost-competitive compared to grid power as can be seen in the energy transition scenarios, it is only pertinent for Nigerian energy planners to exploit the potential of small hydro in the country. It can provide cheaper and low carbon electricity to those communities where the resource is available.

6. Conclusions

In spite of the fact that most residential energy system models built are for developed countries, in this paper, an attempt has been made to develop a dedicated modelling framework for the residential sector of one of the countries in the least developed region of the world. The model has been employed to investigate the least-cost energy system configuration of the targets of the SDG 7 and to quantify the impacts of these targets on environmental emissions. One of the key benefits of the model is the detailed representation of rural and urban households. This disaggregation enabled us to understand the impacts of residential energy transitions by household categories. The developed bottom-up model shows that technologies will play a key role in reducing the energy demand and decarbonisation of the residential sector. The model gives insights on the impact of urbanisation and end-use functions on energy demand which are usually missing in aggregated models. The scenarios analysed in this study were able to identify key interventions for reducing household energy demand and its carbon footprint. Accounting for the externalities cost of fuel use shows that households need to start making rational choices when selecting cooking fuels. Households need to consider co-benefits such as health implications of fuel choices. Finally, it is worth mentioning that this study is part of ongoing work to develop a TIMES model for the entire energy system of Nigeria.

Declaration of competing interest

The authors declare no conflict of interest.

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Appendix A. Techno-economic data of technologies

End Use	Technology Name	Fuel input	Efficiency*	Investment Cost (mill. USD/PJ)	Fixed O&M costs (mill. USD/PJ.a)	Life Time (Years)
Lighting	GLS	Electricity	1.00	0.02	N/A	1
	F-Tube	Electricity	7.00	0.31	N/A	3
	CFL	Electricity	4.90	0.2	N/A	5
	LED	Electricity	9.80	0.54	N/A	10
	Hurricane lamp	Kerosene	0.11	0.21	N/A	3
Cooking	Wick lamp	Kerosene	0.11	0.21	N/A	3
	Induction Stove	Electricity	0.95	44.17	2.21	10
	Kerosene Stove	Kerosene	0.55	4.17	0.21	4
	3-stone Stove	Wood	0.08	0	0	1
	Improved Wood Stove	Wood	0.25	0.33	0.02	2
	Biogas Stove	Biogas	0.55	25.07	1.25	4
	Solar Cooker	Solar	1.00	6	0.30	8
	LPG Stove	LPG	0.60	35.83	1.79	7
	Standard AC	Electricity	1.00	35	1.75	14
Air conditioning	Efficient AC	Electricity	1.30	42	2.10	14
	Super-efficient AC	Electricity	1.50	53	2.65	14
	Standard Geyser	Electricity	0.70	25	1.25	13
Water heating	Efficient Geyser	Electricity	0.90	27	1.35	13
	Super-efficient Geyser	Electricity	1.00	30	1.50	13
	Kerosene Stove	Kerosene	0.55	4.17	0.21	4
	3-stone Stove	Wood	0.08	0	0	1
	LPG Stove	LPG	0.60	35.83	1.79	7
	SWH	Solar	1.00	101	5.05	20
	Standard Fridge	Electricity	0.70	15	0.75	15
Refrigeration	Efficient Fridge	Electricity	0.90	19	0.95	15
	Super-efficient Fridge	Electricity	1.20	21	1.05	15
	Standard Deep Freezer	Electricity	0.80	19	0.95	15
	Efficient Deep Freezer	Electricity	1.10	22	1.10	15
	Super-efficient Deep Freezer	Electricity	1.20	23	1.15	15
Other Electric Appliances	Aggregated Technology	Electricity	1.00	N/A	N/A	N/A

*Figures greater than 1 indicate abstract efficiency.

N/A = Not applicable.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rser.2019.109510>.

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